



Challenges and Perspectives of Interacting with Hierarchy Visualizations on Large-Scale Displays

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ABSTRACT

In this paper we take a look into typical interaction scenarios when it comes to visualizing hierarchical data while we particularly focus on one specific candidate that is the visual depiction by generalized Pythagoras trees. Those are visualizations that benefit from their aesthetically appealing appearance since they come close to a visual model of trees growing in nature. Traditionally, those generalized Pythagoras trees are visualized on typical computer screens, clearly indicating the hierarchy, but negatively, the finer substructures are oftentimes not visible anymore due to the display space restriction and the low resolution. Hence, in this paper we describe the generalized Pythagoras trees displayed on large-scale high-resolution displays and illustrate the challenges and perspectives when we have to interact with the visual representation by using the technique individually, but also in collaboration. To explore possible interactions we experimented with tablet computers combined with an optical tracking system. We evaluate the effectiveness and aesthetics by means of several modifiable visual parameters and by interacting with the visualization.

CCS CONCEPTS

• Human-centered computing → Visualization techniques;

KEYWORDS

Interaction, Large-scale displays, Hierarchy visualization

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1 INTRODUCTION

Various visualizations have been designed over the years focusing on providing perspectives on one or more aspects of a certain dataset of a particular kind of data type or a combination thereof.

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With the advent of the modern computer, those diagrammatic representations [24] became more and more complicated, visually depicting larger and larger datasets, and are equipped with a wealth of interaction techniques supporting a viewer to navigate in the visualization.

But as a drawback of those traditional visualizations we identify the rendering area on which a diagram is visually mapped. In particular, for hierarchical structures that recursively use the same visual metaphor to visually depict finer and finer substructures, the visualizations on traditional computer screens suffer from aggregation, overplotting, and hence, tiny visual structures are not observable anymore. This is in particular the case for hierarchy visualizations [28] using node-link diagram-like representations [6] in which the deeper levels are mostly rendered to smaller areas and hence, the display area rapidly decreases for displaying the hierarchy substructures. Moreover, visual clutter can become a problem if too many elements are displayed in a small display region or if those are not arranged in a suitable layout [26]. For hierarchies this can become a tedious problem if the visual metaphor is based on node-link diagrams or variations thereof and in particular, if node-link representations are used that allow link crossings.

To circumvent some of the problems that traditional computer screens have to some extent, we investigated the idea of visualizing hierarchical data structures on a large-scale high-resolution display, where the human spectator can freely walk around and have a look at the visualized hierarchy from different perspectives and also from different varying distances. To reach this goal we experimented with the generalized Pythagoras trees [1, 2], a visual enhancement to traditional Pythagoras trees that supports the representation of arbitrary trees like file systems, evolutionary trees, or software system architectures. Those trees benefit from their aesthetically appealing appearance and are a good candidate to experiment with typical user interactions.

Consequently, we do not only graphically depict the hierarchy data in form of a generalized Pythagoras tree but we also allow interactions with the large-scale display, i.e., the viewer can inspect certain regions in more detail while also several visual parameters can be changed leading to totally different views on the same hierarchy. Moreover, users are equipped with tablet computers combined with an optical tracking system with which traditional individual interactions but also collaborative ones can be researched for their utility, effectiveness, and intuitiveness. The major goal of this work is to experiment with the visualizations, the interactions, but also with the interplay of the human user with a large-scale high-resolution display. We chose generalized Pythagoras trees as

a candidate of an information visualization technique since it is easy to implement and due to the fact that it generates aesthetically appealing tree diagrams providing really fine structures and substructures.

2 RELATED WORK

There exist lots of visualization techniques for hierarchy data [28], sometimes they are based on 2D, 3D, or hybrid concepts, sometimes they use Cartesian or radial visualizations [4]. Moreover, in most of the cases additional attributes are attached [3]. However, most of them are represented on traditional computer screens, probably suffering from data aggregation and hence, a loss of visual details. This problem is most important for fractal-like [20] and node-link diagram-based [6] hierarchy structures to which generalized Pythagoras trees [1, 2] also belong. The challenging issue with such visualizations is that the finer visual structures in deeper hierarchy levels are not observable anymore if a low display resolution is used and the display space is restricted as in traditional computer screens.

In this paper we chose generalized Pythagoras trees as one method to depict hierarchies since they benefit from their aesthetic appearance due to the fact that they come close to trees growing in nature [23]. They also build self-similar structures and substructures making them some kind of fractal-like representations that themselves have a high degree of aesthetics [19]. Moreover, with the generalized Pythagoras trees, several visual parameters like sizes of angles, the hierarchy traversal or order, the color codings, or the extension from squares to rectangular shapes, are modifiable, resulting in different outcomes of the visualization algorithm and hence, a lot of parameters to experiment with. These aspects make generalized Pythagoras trees a candidate for computational aesthetics. Moreover, by the recursive algorithmic structures, the generated algorithm becomes rather simple to implement, understand, and to illustrate step-by-step to a novice user.

Other techniques for hierarchies rather focus on a nesting of the subhierarchies like treemaps [17], a stacking approach like layered icicles [18, 30], a node-link metaphor showing the parent-child relations more explicitly [25], or the concept of indentation like the indented pixel tree plots [8, 9]. Most of those techniques rather focus on visual scalability, but as a negative consequence, they do not reflect the hierarchy in a natural and aesthetically appealing way like generalized Pythagoras trees [1, 2] do. For example, Burch and Weiskopf [12] investigated the idea of using rapidly-exploring random trees (RRTs) for visualizing large hierarchies as a node-link diagram in a space-filling way, but negatively, their approach cannot display real-world data structures, but rather randomly generated trees. Similar aspects hold for the bubble hierarchies [15] that benefit from their curved bubble shapes, but they also do not depict real-world hierarchies but also random trees. In this work, we focus on real-world datasets to depict hierarchies and investigate their depiction on large-scale displays combined with possible interaction techniques.

As a drawback of the visual mapping of those generalized Pythagoras trees to large-scale high-resolution displays we detected the missing interactions which will be discussed in this work. Moreover, since the display is large enough to allow walking around for

exploring the hierarchy data, those interactions must be adapted to this new hierarchy visualization scenario, hence, we also equipped users with tablet computers to select portions of the generalized Pythagoras tree that they can either explore individually or share with others. Generally, for more than two decades, researchers investigate the applicability of large-scale displays like the CAVE [13] and benefit from the ever growing number of pixels. In the beginning, video projectors build the basis for this new technology [14]. Later on, displays involving flat panels were more common [21] which were advantageous due to their low costs. Papadopoulos et al. [22] researched the Reality Deck using more than 1 and a half billion pixels by exploiting several hundred liquid-crystal displays.

Schmauder et al. [27] already described a distributed visual analytics system briefly exploring tree-like structures on a powerwall, but although they also describe general Pythagoras trees as one visualization candidate, they do not primarily focus on interactions as we do in this work. Moreover, they mostly describe the technical issues that have to be solved for designing a visual analytics system on a large-scale display. Traditional interactions typically applied at a desktop, laptop, or small-scale display like a mobile [29, 32] have to be extended to the new large-scale display scenario. For example, collaborative interaction [16, 31] can be such a scenario which asks questions about if users are rather working synchronously or asynchronously with the visualization or if they are located at the same position (co-located) or not (distributed). In our scenario the users work in a co-located scenario, synchronously with the global overview, but asynchronously with each tablet computer.

3 GENERALIZED PYTHAGORAS TREES

In this section we describe how hierarchy data is mathematically modeled (Section 3.1) and how the generalized Pythagoras trees are designed from it to be representable on traditional computer screens (Section 3.2). As a next step, we illustrate the mapping of the data to large-scale high-resolution displays (Section 3.3) and finally, we demonstrate the impact of parameter changes on the visual appearance of the generalized Pythagoras trees (Section 3.4).

3.1 Hierarchical Data

We model a hierarchy

$$H = (V, E)$$

as a pair of two sets which are the vertices

$$V := \{v_1, \dots, v_n\}$$

and the edges

$$E := \{e_1, \dots, e_m\}$$

that express the parent-child relationships.

Each vertex $v \in V$ is assigned a weight $w_v \in \mathbb{R}_0^+$ that can have two different forms. Either it is only existent for the leaves of the hierarchy, but can be accumulated in the hierarchy up to the root, resulting in a **containment** hierarchy (like in file systems), or each vertex has its own weight (like in phylogenetic trees) resulting in a hierarchy based on the principle of **subordination**. Both weight scenarios are visualizable with generalized Pythagoras trees.

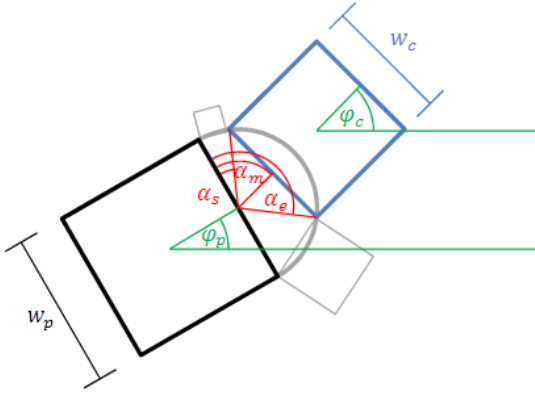


Figure 1: The relevant angles (red color) for the computation of the geometry of a child node (blue color) with respect to its parent node (black color). This tree is splitting into three subtrees at the black colored node.

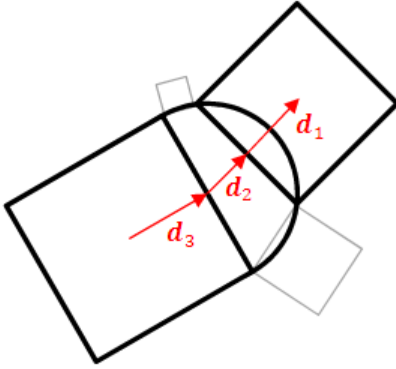


Figure 2: Vectors d_1 , d_2 , and d_3 for computing the center points of the child nodes with respect to the parent node size. This tree is branching into three subtrees at the parent node.

If a hierarchy H is branching at a certain vertex $u \in V$ we denote the corresponding subhierarchy by H_u indicating that its root vertex is $u \in V$. This subhierarchy may consist of a set of $m \in \mathbb{N}$ subsubhierarchies denoted by H_{u_i} , $i \leq m$.

3.2 Visual Design

Visually mapping a hierarchical structure to graphical primitives can be done in an easy way since those structures consist of self-similar recursive substructures. In particular, for the visualization with generalized Pythagoras trees we only have to understand how the branching effect is computed and visually represented. This is demonstrated in Figure 1 for the angles and in Figure 2 for the center points for a child node with respect to a parent node.

Figure 1 shows how the branching is computed based on a parent node. The deepest parent node in a hierarchy is the root node which is placed constantly with a center point $(0, 0)$. The side lengths are both mapped to 1, i.e., the basic shape is a square. Depending on the child nodes in a subhierarchy the angles on a half circle are computed which give a total value of 180 degrees. Mapping larger subhierarchies to larger angles and smaller ones to smaller angles results in a weight-preserving hierarchical representation by a generalized Pythagoras tree.

In this visualization step there are already several parameters that can be modified. For example, the angle sizes can be changed based on a weight function or another kind of hierarchy-specific metric. Moreover, the traversal of the hierarchy can play a role about the visual appearance, i.e., if the tree looks more balanced or not, if for example, the child nodes for each parent node are displayed in an order that places large ones closer to the center of the half circle. Also the color coding is changeable that can indicate different categories of nodes like, for example, in a file system. Finally, also the squares might be changeable to rectangular shapes allowing to reduce the node overlap problem a bit by moving certain branches further apart from the corresponding parent nodes which has already been discussed in the work of Beck et al. [1, 2].

3.3 Mapping to Large-Scale Displays

The general idea of mapping hierarchical data to a generalized Pythagoras tree is rather simple, if the goal is to not compute an overlapping-free hierarchy visualization (which could be solved by a variant of a force-directed layout being algorithmically more complex). The overlaps are not that relevant since trees in nature, if inspected from far away, also do not provide a visual appearance that is free of overlaps. However, they are nonetheless aesthetically appealing natural structures and patterns, in particular, if they are enhanced by beautiful colors like in autumn time. Hence, the focus of this work is not primarily on reducing those visual overdraw effects, but we rather focus on providing a large-scale view on the hierarchy, also reflecting finer substructures that would not have been visible if rendered on a traditional computer screen. Figure 3 illustrates a powerwall display that we used for rendering the generalized Pythagoras tree while the width and height are given in meters and pixels.

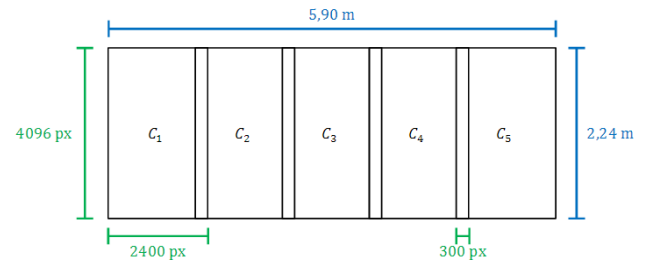


Figure 3: The large-scale display with meter and pixel annotations indicating width and height of the display.

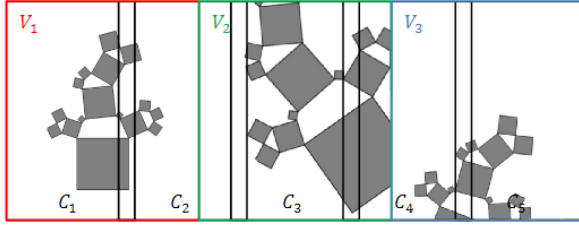


Figure 4: Three users are interacting with the powerwall requiring a subdivision into subareas on the large-scale display. At the borders the individual regions from a generalized Pythagoras tree are cut to avoid overlaps.

The display is a powerwall of 5.90 meters $\times$ 2.24 meters with a rear projection system. The total resolution of it is 10,800 $\times$ 4,096 pixels which is obtained by five projectors of the type DLA-SH4K '4K' by JVC having a resolution of 2,400 $\times$ 4,096 pixels each. The projectors are operated by a cluster node with two graphics processing units (GPUs). Gigabit Ethernet with IP over InfiniBand is used to achieve a good connection of the cluster nodes.

Since we also support the rendering of several subareas of a generalized Pythagoras tree due to several viewers interacting with the powerwall display and the visualized tree, the display can be subdivided into several areas which is shown in Figure 4. This means each user can get his own workspace in which he can see a part of the generalized Pythagoras tree.

3.4 Modifiable Visual Parameters

As already described in the work of Beck et al. [1, 2] there are several visual variables that might be modified in order to change the visual appearance of the computed generalized Pythagoras tree. For example, the angle subdivision at each halfcircle from 180 degrees into as many subangles of existing branches has an impact on the layout of the tree. These angles can be based on several metrics, depending, for example, on the fact if a containment hierarchy or a subordination hierarchy has to be visualized. In Figure 5 we can see two scenarios in which either equally large square sizes are used for the sibling vertices in a hierarchy (a) or the number of corresponding leaf vertices in each subhierarchy can be taken for the square sizes (b). Those sizes have an impact on the angles and vice versa. Figure 5 also illustrates a scenario with different color codings. In (a) only red color is used whereas in (b) a rainbow colormap is chosen that generates more aesthetically appealing diagrams.

4 INTERACTIONS

A list of interaction techniques typically based on tablet computers (Section 4.1) is considered as useful operations to navigate in the hierarchy and to provide different perspectives on the visualized data. Those can be subdivided into techniques for an individual user (Section 4.2) and those for collaborative interaction (Section 4.3). Moreover, general interaction techniques are described (Section 4.4).

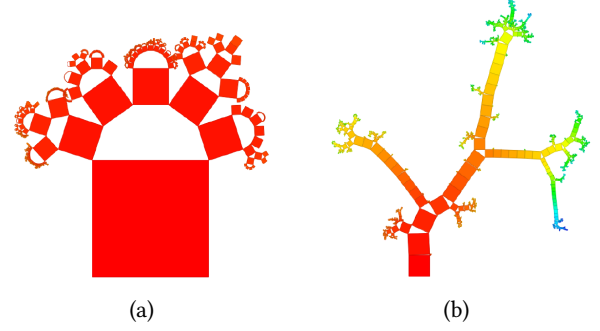


Figure 5: The NCBI taxonomy: (a) Sibling nodes are equally large squares. (b) The size of the squares is dependent on the number of leaf vertices in each subhierarchy.

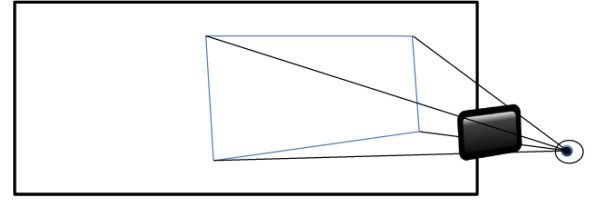


Figure 6: The virtual view pyramid of a user who looks at the powerwall through the tablet computer

4.1 Tablet Computer Interaction

A hybrid form of interaction is implied by an optically tracked tablet computer. This is a fully functionable computer that can be used to execute complex computations on its own for navigating in parts of the generalized Pythagoras tree and to further explore such a hierarchical dataset. Another major benefit comes from the fact that each tablet computer can be connected to all the others providing a way to collaboratively interact with all the other users but also with the powerwall display that serves as some kind of overview and context representation. The tablets and the powerwall become complex overview and detail or focus and context displays.

The basic idea in this work is that the tablet computer can be used as a window through which we can look at certain regions in space (see Figure 6), i.e., the tablet itself provides a frame that describes which part of a larger visualization (shown on a powerwall) is currently selected. The orientation and position of the tablet play a crucial role about which part of the visualization is currently under exploration.

The tablet can also be used to look at the visualization from different perspectives like general transformations applied to the representation. This can be a translation, a scale, or a rotation around the z-axis. Devices allowing multi-touch interactions are particularly useful for gestures for these transformations. Figure 7 illustrates how these interactions by means of a tablet computer are achieved. By applying this interaction the user can bring far away places in the visualization in a reachable and visible region on a traditional computer screen provided by the tablet computer.

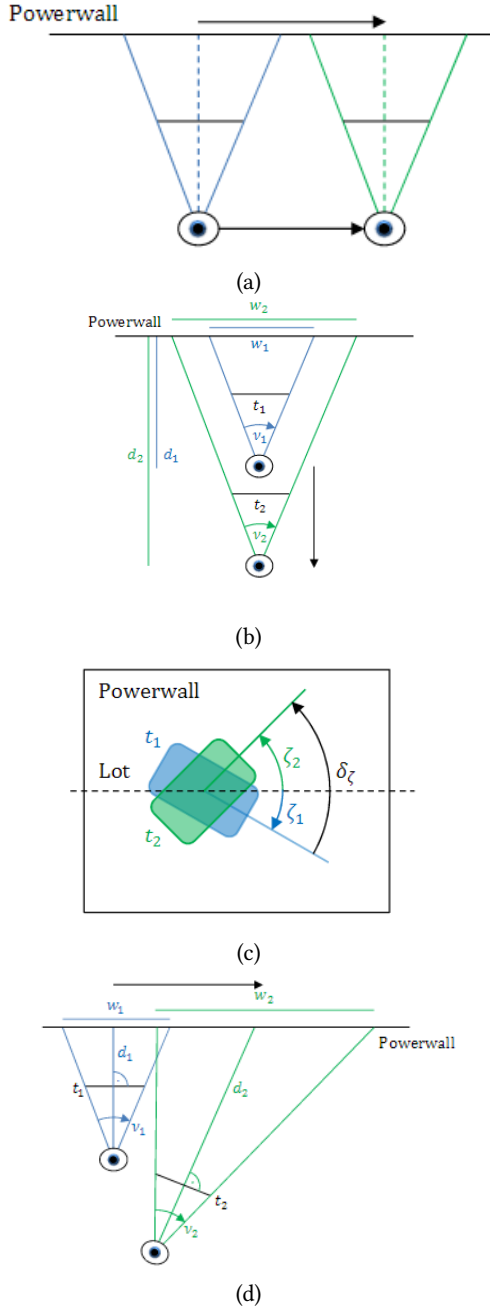


Figure 7: Possible transformations of the view by interaction techniques which are achieved by means of the optical tracking system: (a) Translation by moving the tablet computer horizontally as well as vertically. (b) Scaling by moving to the powerwall or away from it. (c) Rotation around the z-axis by rotating the tablet computer. (d) Translation and scaling by using different degrees of freedom.

A selection interaction can, for example, be used to further inspect certain nodes in the generalized Pythagoras tree. This is

shown in Figure 8. The user focuses a region on the powerwall in which an interesting node is located. If it is visible on the tablet computer, it can be directly selected by touch interaction on the tablet computer. Then an additional information about the selected node can be displayed on the tablet display which does not clutter the visualization for all the other users.

4.2 Individual User Interaction

By using gestures, the visualization shown on the large-scale display could be modified. Figure 9 shows simple individual user interactions which can be described as translation, scale, and rotation but also selections and highlighting of hierarchy elements is possible. Moreover, details-on-demand and the expansion and collapse of subhierarchies is supported.

The transformations (Figure 9 (a)) can be achieved by gestures that change the view directly on the powerwall. In the selection interaction (Figure 9 (b)) we can use the tablet by doing a tap gesture that leads to a highlighting of the nodes in the hierarchy. The details-on-demand interaction (Figure 9 (c)) works by adding a two-dimensional overlay on top of the rendered generalized Pythagoras tree. Finally, the expand and collapse interaction (Figure 9 (d)) demands for selecting a certain parent node or already collapsed node on the display.

In particular, the collapse interaction can be designed in two ways as shown in Figure 10, i.e., by either removing the collapsed subtree while leaving the generalized Pythagoras tree layout as it is (Figure 10 (b)) or by removing the collapsed subhierarchy while computing a new (more balanced) layout for the generalized Pythagoras tree (Figure 10 (c)).

4.3 Collaborative Interaction

A large-scale display can be used by multiple users since they have enough space to freely walk around and explore an individual subarea of the display, i.e., a certain part of the displayed generalized Pythagoras tree. To reach this goal, the users were equipped with tablet computers, which are important means to interact with the display and the visualization.

Figure 11 illustrates several of those interactions that also provide highlights of elements selected by other users which is in particular useful to have an additional context information about the visual elements that are under exploration by the other users.

4.4 General Interactions

Apart from individual interactions with a tablet computer or collaborative interactions combining several users explorations, there are also general visualization-specific interaction techniques. For example, following traditional interaction design taxonomies [32] we can implement an interaction that highlights a certain path in a hierarchy starting at the root node and leading to the searched leaf node. Such a scenario is illustrated in Figure 12 in which the leaf path for the homo sapiens is highlighted in the generalized Pythagoras tree depicting the NCBI taxonomy.

As we can see in Figure 12 the tree structures becomes rather fine-structured in deeper levels and although it can be displayed

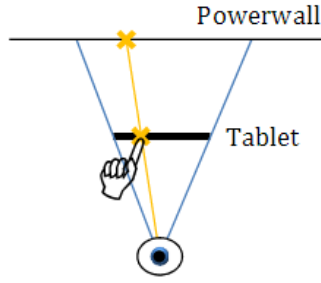


Figure 8: Selection of a node at the powerwall by applying the touch interaction of the tablet computer

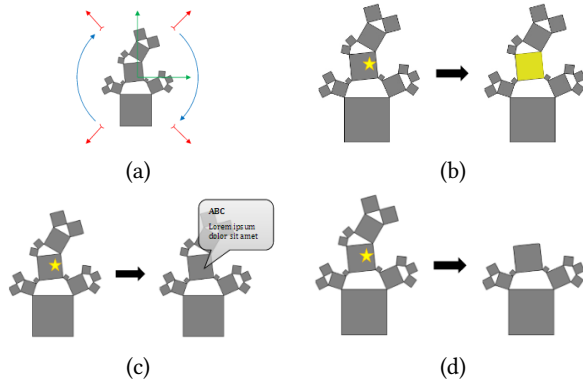


Figure 9: An overview about interaction techniques for an individual user: (a) Translation, scale, and rotation of the view. (b) Selection and highlighting of hierarchy elements. (c) Details-on-demand for hierarchy elements. (d) Expanding and collapsing subhierarchies.

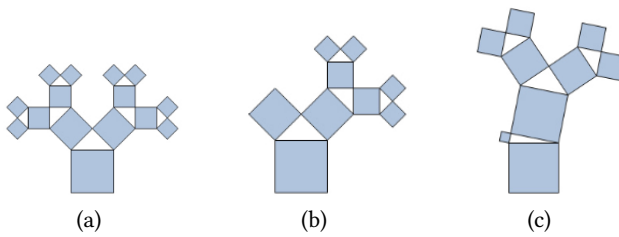


Figure 10: Expanding and collapsing a subhierarchy in a generalized Pythagoras tree: (a) The original tree. (b) Collapsing of a subtree by removing the subtree while the entire tree layout is not changed. (c) Collapsing of a subtree by removing the subtree while the entire generalized Pythagoras tree is redrawn in a new layout.

at the large-scale display, a user might be interested in these fine substructures in the deepest levels of the hierarchy. For this reason, we can also apply a geometric zoom technique that is illustrated in Figure 13.

5 APPLICATION EXAMPLE

We tested the system with different hierarchy datasets to get an impression about scalability aspects, i.e., how large and how deep hierarchical structures can become to still reliably work with them on a large-scale high-resolution display and also with respect to the interaction techniques. Figure 14 illustrates two datasets and the generalized Pythagoras trees by visualizing the directory/file sizes and the sizes of subtrees, respectively, for a file system and for a phylogenetic tree.

As we can see from the figure, there are various fine substructures indicating the subtrees of the generalized Pythagoras trees. Moreover, the file system tree contains many elements on the lowest level (the files) while the phylogenetic tree seems to be more deeply structured than the file system.

The phylogenetic tree consists of 324,276 elements at a maximal depth of 42. There are 263,691 leaf nodes that each represent an individual species. The phylogenetic tree is displayed on the large-scale display while a person is standing in front of it, using a tablet to interact with the tree and with parts of it (Figure 15 (a)). The user has a closer look at an individual branch of the generalized Pythagoras tree which stands for a splitting off from a certain kind of species indicated by the corresponding root node. On the tablet the user can see additional information about the tree branch in focus, e.g., the name of the species or a suitable and representative image for the species indicated by the parent node. Moreover, the user can highlight and store the just explored visual element, either for later explorations or for other users, i.e., it can be marked as important or worth investigating. The marked elements can be requested by the other viewers and they can decide how to proceed with the marked information. In Figure 15 (b) the user selected a part of the generalized Pythagoras tree that seems to be worth investigating.

6 CONCLUSION AND FUTURE WORK

In this paper we described a way to visually encode hierarchical structures based on generalized Pythagoras trees that benefit from their aesthetical appearance due to the similarities to trees growing in nature. To also illustrate how the finer granular hierarchical substructures can be made visible we investigated a system to visually encode them on large-scale high-resolution displays. To fully benefit from those powerwall displays we discuss interaction concepts with which those generalized Pythagoras tree visualizations become explorable. For future work we plan to evaluate our powerwall visualization by experts from the field of information visualization. Moreover, we plan to visually map further information visualizations to such a large-scale high-resolution display in order to not only experiment with a single visualization technique, but to try to understand for which other visualization techniques such large-scale displays are particularly useful. For example, dynamic graph visualizations [5, 7, 10] or representations of general time-varying data [11] might be worth investigating since those have to be scalable in many data dimensions, where time is the most crucial one.

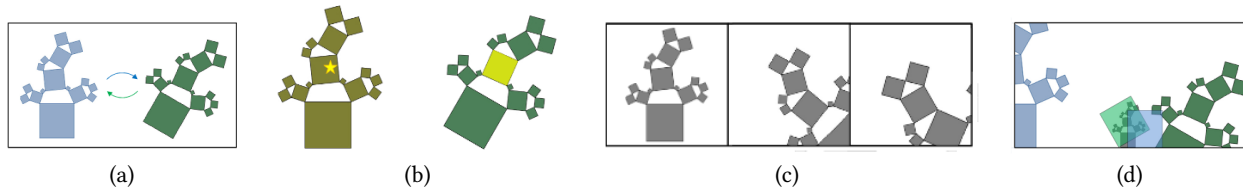


Figure 11: Collaborative interaction techniques: (a) Change of user perspectives. (b) Showing highlights of other viewers in the own view. (c) Snapshots to undo interactions. (d) Overview with subareas from the display between neighbored views.

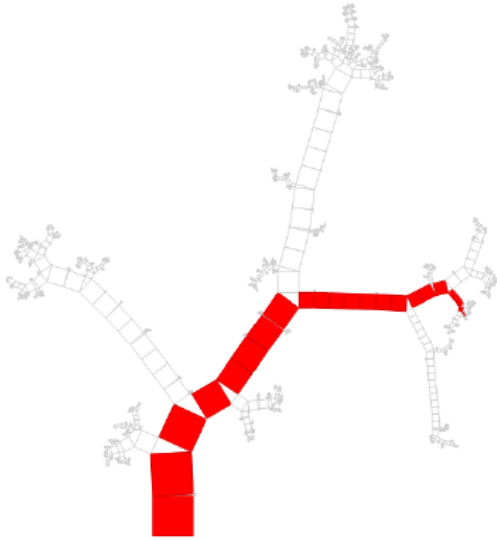


Figure 12: The path to the species homo sapiens is highlighted and the other nodes are grayed out.

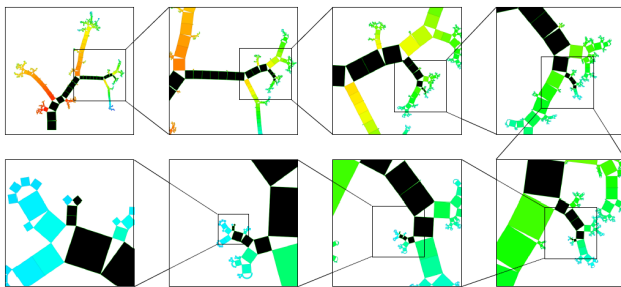


Figure 13: Geometric zoom along a path in the phylogenetic tree to the species homo sapiens

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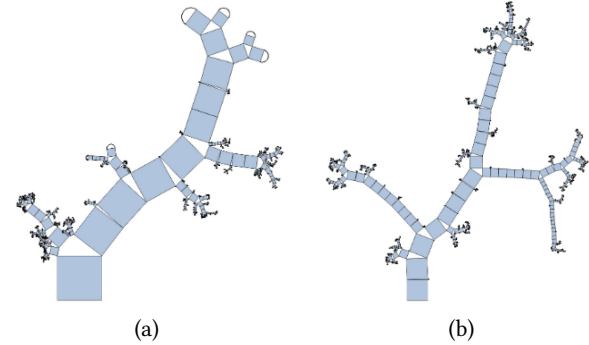


Figure 14: Generalized Pythagoras trees for two application examples: (a) A file system. (b) A phylogenetic tree.

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(a)



(b)

Figure 15: A generalized Pythagoras tree on a large-scale high-resolution display: (a) In the initial state the entire dataset is visualized, in this case twice since two individual users experiment with the visualization. (b) The user on the left hand side has zoomed a certain region in the generalized Pythagoras tree to explore it in more detail and on a finer-granular view.

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